YASH SINHA

AI/ML Engineer • Data Scientist

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PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with 6+ years of applied research experience building and deploying production-grade machine learning systems. Proven expertise in end-to-end ML pipelines — from data ingestion and feature engineering to model training, evaluation, and cloud deployment. Hands-on track record across LLM-driven agents, computer vision/biometrics, quantitative finance, time-series forecasting, and NLP. Inventor of patent-pending AI/ML methodologies. Adept at translating complex research into scalable, real-world solutions on AWS and other modern infrastructure.

TECHNICAL SKILLS

Languages:	Python, SQL, R, C/C++, JavaScript, Java, Bash, MATLAB, Perl
ML/AI Frameworks:	TensorFlow, PyTorch, Keras, Scikit-learn, LangChain, LangGraph, LangSmith, Ollama, LlamaIndex, OpenAI API, Anthropic API
Cloud, Data, & MLOps:	AWS (EC2, S3, RDS, Lambda, API Gateway, SageMaker, Glue, Athena, IAM, CloudWatch), Databricks (Apache Spark, Delta Lake, MLflow, Unity Catalog, Databricks SQL), Terraform, Docker, Git/GitHub, Linux, CI/CD Pipelines, REST APIs, Apache Airflow
Domains:	LLMs, AI Agents, RAG Systems, ML, NLP, Quantitative Finance, Financial Forecasting, Time Series Forecasting, Statistical Modeling, ETL/ELT Automation, Data Quality Automation
Certifications:	CFA Level I Candidate (Nov 2026) • AWS Certified Solutions Architect • CompTIA Linux+

PROFESSIONAL EXPERIENCE

Chief AI/ML Engineer & Chief Data Scientist | PropertyDealz LLC (Startup) *June 2025 – Present*

AI-Powered Real Estate Investment Analytics Platform

Co-founded and led ML engineering and data science for an AI-driven real estate platform that identifies undervalued properties and pricing anomalies. Architected end-to-end pipelines combining advanced statistical models, quantitative signals, natural language property descriptions, and geospatial neighborhood data.

- Designed and deployed advanced statistical models and ensemble ML algorithms for property price forecasting and anomaly detection, integrating tax history, price-per-sq-ft trends, and foreclosure signals to surface high-confidence investment opportunities.
- Designed and deployed multi-agent AI workflows using LangGraph, LangChain, OpenAI APIs, and RAG architectures to automate property intelligence gathering and generate structured investment reports.
- Engineered geospatial scoring systems ingesting neighborhood signals from social media and public data APIs, improving investment signal precision.
- Developed scalable ETL/ELT pipelines using PostgreSQL, AWS Lambda, S3, and Glue to ingest and transform MLS datasets while reducing latency through event-driven architectures.
- Implemented Infrastructure-as-Code with Terraform and deployed production ML workloads across AWS (SageMaker, EC2, Lambda, API Gateway, RDS, S3, Glue) supporting real-time inference and automated workflows.

ML Research Engineer (Independent) | Quantitative Trading Research *2025*

Reinforcement Learning for Stock Trend Prediction via LLM-Augmented News Ingestion

Designed and implemented a research prototype for automated trading signal generation, combining LLM-based news analysis with RL-optimized decision making.

- Engineered a full NLP pipeline using LLMs to summarize financial news, extract named stock tickers, classify market sentiment, and surface buy/sell/hold signals.
- Applied reinforcement learning (return-driven reward shaping) to optimize trade action policies against historical market data.

- Integrated RAG (Retrieval-Augmented Generation) to ground LLM recommendations in relevant historical context, improving signal quality for unseen news.
- Orchestrated end-to-end daily workflows (ingestion → LLM inference → RL update → storage) using Apache Airflow with CI/CD integration.

Lead Research Assistant | SMU AT&T Center for Virtualization (SRC/Intel Sponsored) *Aug 2023 – Jul 2025*

ML-Aided Power Obfuscation for Cryptographic Hardware Security

Led a five-person cross-disciplinary team developing latch-based circuit modifications to defend against power side-channel attacks. Applied ML techniques to evaluate and improve circuit security.

- Developed novel mux-latch variants of DES56 encryption; used Cadence Joules to simulate and compare power traces from VCD files.
- Applied ML-based TVLA (Test Vector Leakage Assessment) to statistically quantify a circuit's vulnerability to power analysis attacks, informing iterative design improvements.
- Evaluated ML models trained on power trace data to benchmark resistance across DES, AES, and ASCON implementations.
- Presented research at SRC TechCon 2024 and 2025; work published in two conference proceedings.

Research Assistant | SMU Darwin Deason Institute for Cyber Security (Raytheon Sponsored)
Jun 2022 – Jun 2023

Biometric Authentication via Applied Machine Learning

Built an Android-based two-factor biometric authentication system leveraging on-device ML inference and cloud-backed model retraining.

- Designed end-to-end ML pipelines including feature engineering, preprocessing, model training, hyperparameter tuning, and evaluation using TensorFlow/Keras for biometric authentication.
- Implemented on-device model retraining (TFLite) to enroll new users with minimal overhead, enabling continuous learning at the edge.
- Built cloud-native analytics infrastructure using AWS Amplify, Lambda, and S3 for telemetry collection, model versioning, and inference monitoring.
- Coordinated daily standups with Raytheon sponsors; led QA testing and iterative model improvements based on sponsor feedback.

EDUCATION

Southern Methodist University *Dallas, TX*

Ph.D. Computer Engineering *(On Leave)* 2025 – Present

Research Focus: Hardware Security and Machine Learning — ML-driven side-channel analysis, hardware Trojan detection, and AI-assisted security evaluation of cryptographic and edge computing systems.

M.S. Computer Engineering Spring 2025

B.S. Computer Science (Specialization: AI & ML) • B.S. Statistical Science Spring 2023

PUBLICATIONS & PRESENTATIONS

- ***AI-Powered Real Estate Investment Analytics and Property Valuation Algorithms — Patent Pending***
  Developed proprietary ML-driven methodologies and automated decision pipelines for identifying undervalued real estate assets using multi-source market, geospatial, and behavioral data signals.
- ***ML-Based Analysis of Power Trace Resistant Circuit Optimizations — SRC TechCon 2025***
  Evaluated ML-driven security benchmarks across DES, AES, and ASCON cryptographic implementations.
- ***Circuit Optimizations for Power Trace Modification and Attack Resistance — SRC TechCon 2024***
  Developed latch-based designs to harden DES and ASCON against power side-channel attacks.
- ***Real-Time Situation Awareness Assessment for Pilots via Machine Learning — MODSIM 2022***
  Classified pilot cognitive/psychological state during flight maneuvers using ML models trained on sensor data.
- ***Quantum-Assisted Circuit Partitioning for Optimized VLSI Testing — Ph.D. Research, SMU (2025)***
  Developed hybrid quantum-classical design-for-test (DFT) methodologies leveraging quantum circuit partitioning and classical optimization algorithms to improve scalable fault testing, test pattern efficiency, and hardware validation workflows for complex integrated circuits.